

University of Cape Town
Department of Electrical Engineering
EEE3094S - Control Systems Engineering

Problem Set 7.1 Design Specifications

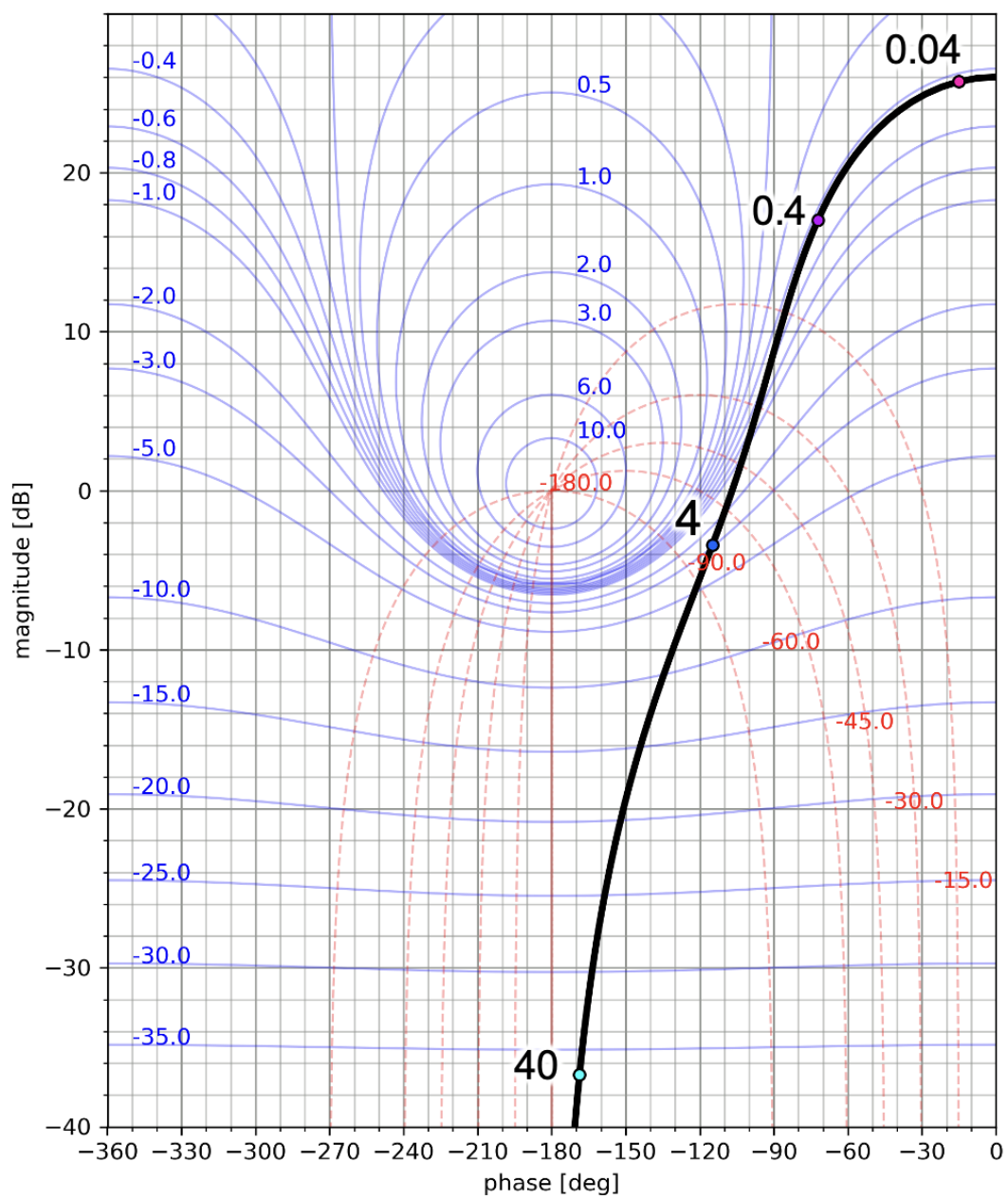
Question 1

The time response of a closed-loop, first-order system must satisfy the following design specifications. Rewrite each of them in terms of frequency-domain parameters:

- A) Tracks a position setpoint with $< 5\%$ error.
- B) Settles to within 5% of its final value < 3 seconds

Question 2

For some design, it is decided that the complex pair $-4 \pm j3$ would be an acceptable target for the closed-loop system poles. The open-loop gain must be at least 15 to satisfy the steady-state error requirements. Based on the Nichols chart (of the open-loop frequency response) decide whether the system is acceptable.



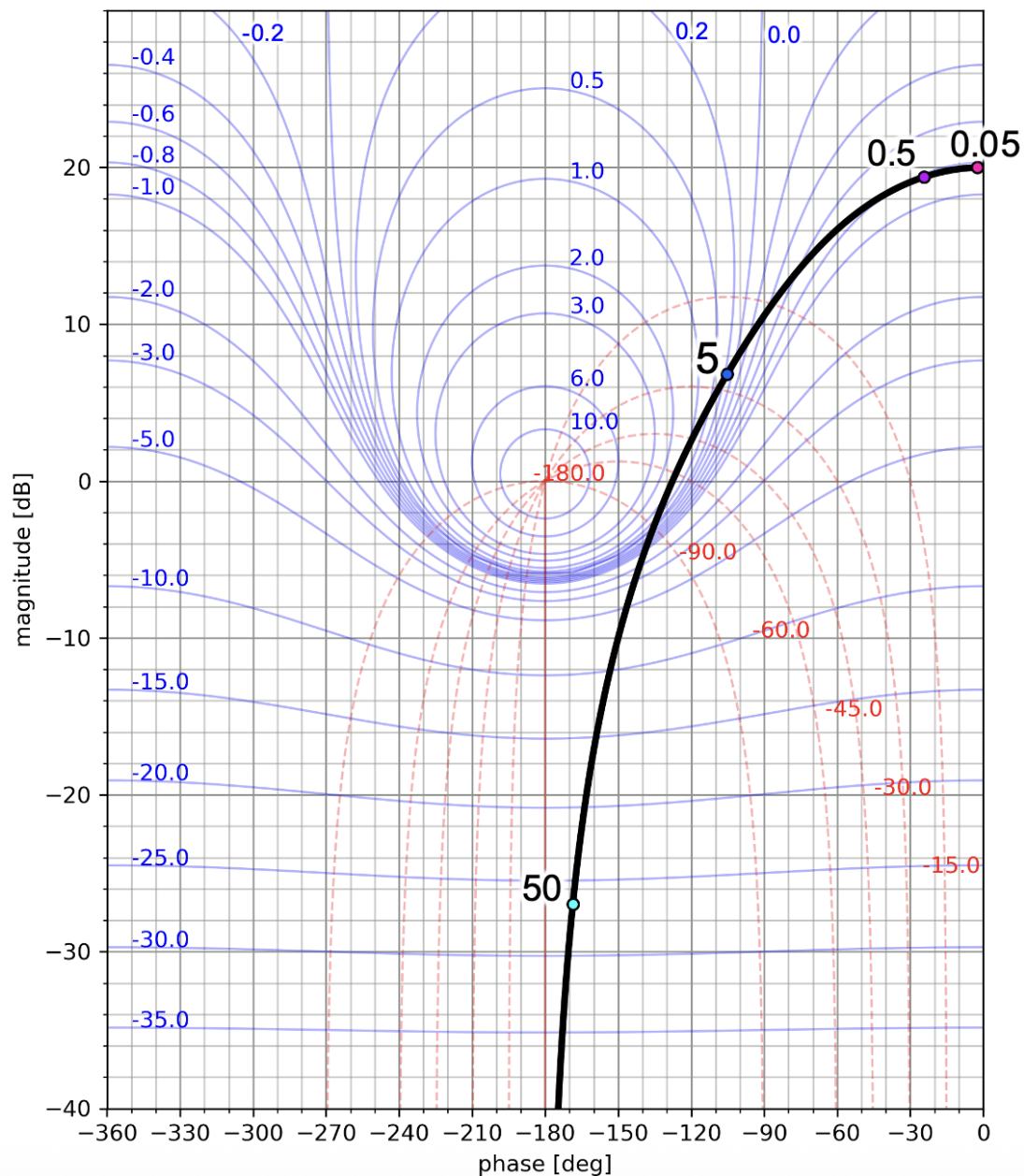
Question 3

A closed-loop system is expected to satisfy the following performance requirements:

- Settling time to 2% of final value of < 1 second
- Percentage overshoot of $< 10\%$ (you may use the formula: $\%OS = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$)
- Position error of $< 10\%$

A) Draw a region on the s-plane representing the closed-loop pole locations that satisfy these specifications.

B) Decide whether the system in the Nichols chart below satisfies the requirements.



Hints:

The open-loop transfer functions used to generate the Nichols charts in questions 2 and 3 are as follows:

Question 2: $G(s) = \frac{23.81}{s^2 + 8s + 1.19}$

Question 3: $G(s) = \frac{113.6}{s^2 + 10s + 11.6}$

You can use these to check your answers or generate your own plots to help you calibrate your approximations.